

SOFTWARE REQUIREMENTS SPECIFICATION

JOB PREPARATION AI

AI-Powered Resume Analysis, ATS Optimization and Interview Preparation

Document	Details
Version	1.0
Status	Initial Requirement
Product Type	Web Application
Primary User	Job Seeker
Technology Context	MERN Stack with Gemini AI

Client Requirement and SRS — concise functional specification

1. Project Overview

The client requires a web-based Job Preparation AI platform that helps job seekers evaluate and improve their resume for a specific job, check ATS readiness, generate relevant interview preparation, and practice through an AI-powered interview experience.

The system shall provide one connected workflow from Job Description and Resume analysis through resume improvement, ATS evaluation, PDF generation and interview preparation.

2. Project Scope

- User registration, login, logout and forgot/reset password.
- Upload a resume and enter a job description.
- AI-based comparison of the resume with the job description.
- Identification of matched skills, missing skills, partial matches and resume improvement areas.
- AI-assisted resume improvement with user control over changes.
- ATS analysis and score based on the job description and selected resume version.
- Keyword, skills, experience, structure and other relevant ATS checks.
- Generation and download of the latest updated resume in PDF format.
- History of previous job analyses and associated resume versions.
- Technical and behavioral interview preparation based on the target job and resume.
- Basic, Intermediate and Advanced interview difficulty levels.
- Interactive AI mock interview with answer evaluation and feedback.
- Project/application chatbot for user guidance and frequently asked questions.

3. Client Requirements

ID	Requirement	Client Requirement
CR-01	User Account	The user shall be able to register, log in, log out and recover a forgotten password.
CR-02	Job Description	The user shall be able to enter a complete job description and optionally provide job title and company name.
CR-03	Resume Upload	The user shall be able to upload a resume in the supported file format. The initial release shall support PDF.
CR-04	Resume Analysis	The system shall analyze the resume against the entered job description and display the level of alignment.
CR-05	Skill Gap Analysis	The system shall identify relevant matched, partially matched and missing skills based on the job requirements.
CR-06	Resume Improvement	The system shall identify areas of the resume that should be improved for the selected job and provide suggested changes.
CR-07	User Control	AI suggestions shall not automatically become final resume content. The user shall review and accept or reject proposed changes.
CR-09	ATS Analysis	The system shall evaluate the selected resume against the job description and provide an ATS score and supporting

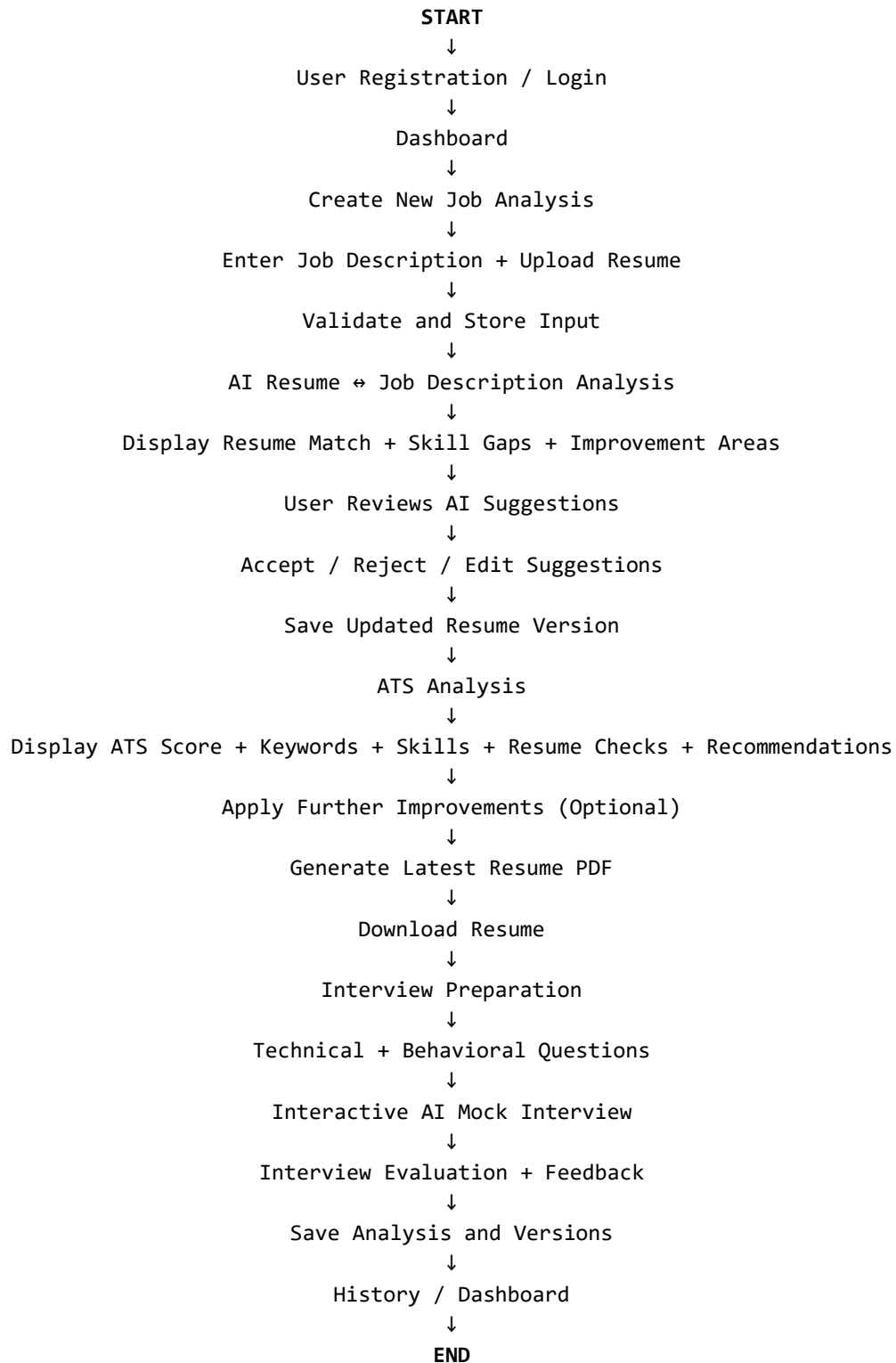
		details.
CR-10	ATS Details	The ATS report shall include relevant keyword alignment, skill alignment, resume structure/sections, experience alignment and prioritized improvement recommendations.
CR-11	ATS Recheck	The user shall be able to re-run the ATS analysis after modifying the resume.
CR-12	PDF Resume	The user shall be able to generate and download the latest accepted resume version as a PDF.
CR-13	Resume Versions	The system shall retain the original resume separately from updated resume versions.
CR-14	Analysis History	The user shall be able to view previous job descriptions, associated resumes, analysis results and latest ATS scores.
CR-15	Previous Analysis	The user shall be able to open a previous analysis and download its latest associated resume.
CR-16	Interview Questions	The system shall generate technical and behavioral interview questions relevant to the entered job description and the user's resume.
CR-17	Difficulty	The interview module shall support Basic, Intermediate and Advanced questions.
CR-18	Additional Questions	The user shall be able to request additional or follow-up interview questions.
CR-19	Interactive Interview	The user shall be able to answer interview questions interactively and receive AI-based feedback.
CR-20	Interview Evaluation	The system shall provide an evaluation of the user's answers, including strengths, weaknesses and areas for improvement.
CR-21	Chatbot	The system shall provide a chatbot for questions about the platform, features and, where applicable, the user's current analysis.

4. AI Requirements

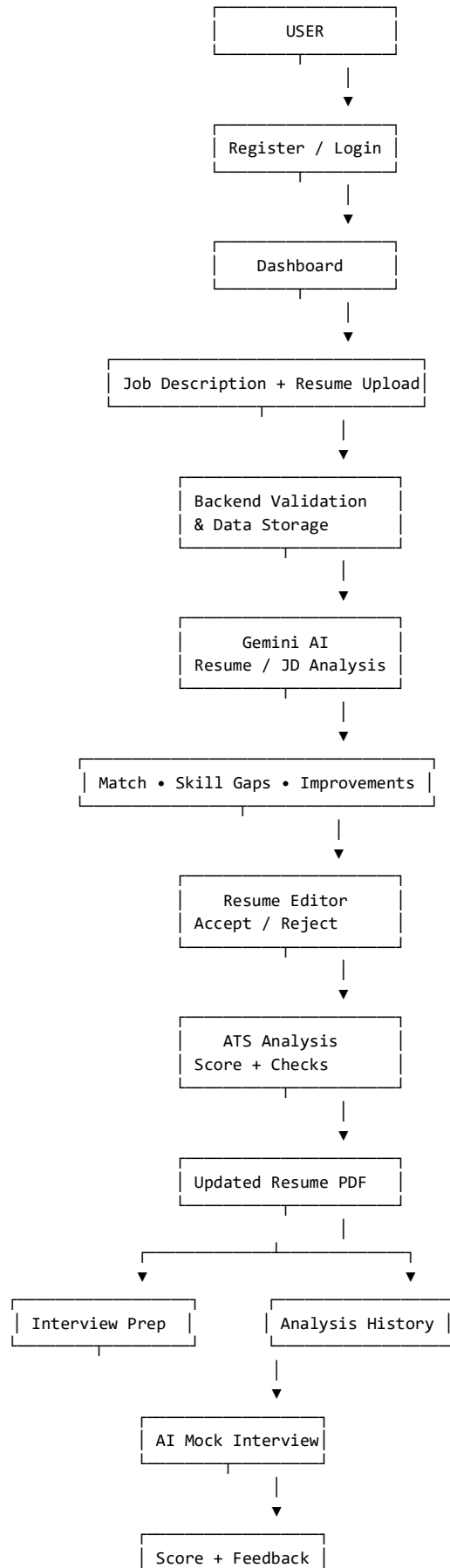
- Gemini shall be used for AI-based resume analysis, semantic job/resume comparison, resume suggestions, interview question generation, interview evaluation and chatbot responses.
- AI output shall be returned in a structured format wherever the application depends on specific fields or scores.
- The application shall validate structured AI output before using it in the user interface or storing it as analysis data.
- AI shall use the job description and resume as the primary context for job-specific analysis.
- AI-generated resume content shall not invent qualifications or professional experience.
- The application shall clearly distinguish user-provided information from AI-generated suggestions.

- ATS scoring shall not depend entirely on an LLM response. Measurable scoring factors should be calculated by application logic, while AI may be used for semantic matching and recommendations.

5. Basic Functional Flow



6. Basic System Flow Diagram



Contextual Chatbot available across the application

7. Required Application Modules / Screens

Screen / Module	Required Functionality
Login / Register	Account creation and authentication.
Forgot / Reset Password	Password recovery.
Dashboard	Summary of analyses, latest ATS score, resume versions and interview readiness.
New Analysis	Job description entry and resume upload.
Resume Analysis	Match result, skill gaps, resume issues and recommendations.
Resume Fix	Review, edit, accept/reject AI suggestions.
ATS Analyzer	ATS score, keywords, skill match, structure and improvement recommendations.
Resume Preview / Download	Preview and generate latest accepted PDF resume.
History	Previous job analyses, resumes, versions, scores and downloads.
Interview Preparation	Technical/behavioral questions with difficulty selection.
Mock Interview	Interactive questions, answers and AI evaluation.
Interview Report	Performance score, strengths, weaknesses and improvement areas.
Chatbot	Platform FAQ and contextual user guidance.

8. Expected Outputs

- Resume-to-job match analysis.
- Matched, partial and missing skills.
- Resume improvement recommendations.
- Updated resume version after user approval.
- ATS score from 0 to 100 with detailed factors.
- Keyword and requirement alignment information.
- Downloadable PDF resume.
- Technical and behavioral interview questions.
- Interactive interview feedback and score.
- Historical analysis and resume versions.
- Context-aware chatbot responses.

9. Basic Data to be Stored

Data	Required Information
User	Name, email, securely stored password information, account timestamps.
Job Analysis	User reference, job title, company, job description, analysis date.
Resume	User reference, original file metadata, parsed resume information and version.
AI Analysis	Match results, skills, gaps, recommendations and analysis status.
ATS Result	Score, keyword results, skill results, resume checks and recommendations.

Resume Version	Version number, content, source analysis and generated PDF reference.
Interview Session	Analysis reference, difficulty, questions, answers, evaluation and score.
Chat Session	User reference, analysis context where applicable, messages and timestamps.

10. Basic Non-Functional Requirements

- Security: Authentication, authorization and secure password storage are required.
- Privacy: User data and uploaded resumes must be isolated by user account.
- File Security: Uploaded files must be validated by type and size before processing.
- Reliability: AI or PDF processing failures must return a clear error and retry option where applicable.
- Performance: Long-running AI operations must display an appropriate loading/processing state.
- Maintainability: Backend business logic shall be separated into modular services.
- Scalability: AI, ATS, resume and interview services should be independently maintainable.
- Responsive UI: The web application should work on desktop and common tablet/mobile screen sizes.

1. Basic Acceptance Criteria

- User can create and access an account.
- User can submit a job description and PDF resume.
- System can produce a structured job/resume analysis.
- System identifies relevant skill gaps and resume improvement areas.
- User can review and control AI-generated resume changes.
- System produces an ATS score and supporting analysis.
- User can generate and download the latest accepted resume as PDF.
- Previous analyses and resume versions can be viewed by the owning user.
- Interview questions are relevant to the selected job and resume.
- Interactive interview provides answer feedback and an overall evaluation.
- Chatbot provides platform-related assistance and respects user data boundaries.

13. End-to-End Product Objective

The final product shall provide a single, connected experience in which a job seeker can evaluate a resume against a target job, understand and address skill and resume gaps, improve ATS readiness, download an updated resume, and prepare for the same job through personalized interview practice.